Compiler

(CS3104)

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Syllabus

1. Introduction to Compilers: The role of language translation in programming; Interpreters Vs. Compilers, Language translation phases, Machine-dependent and machine-independent aspects of translation
2. Lexical Analysis: Regular expressions, Finite automata, Conflict Resolution
3. Syntax Analysis: Context-free grammars, Ambiguous grammers, Top-down parsing (LL parsing), Bottom-up parsing (LR parsing), CYK parser, Conflict resolution
4. Semantic Analysis: Formal Language Semantics, Symbol tables, Type checking, Attribute grammars
5. Intermediate Code Generation: Various Intermediate Representations (IR) of code (e.g., Abstract syntax trees (AST), Single Assignment (SA), Three-address code, etc.), Translation schemes
6. Optimization Techniques: Various optimization scopes (Constant folding, constant propagation, invariant motion, etc.), Data-flow analysis (Liveness, available expression, very busy expression, reaching definition)
7. Code Generation: Instruction selection, Code generation for different architectures; Code generation tools

Books

[1] Aho, A. V., Sethi, R. and Ullman, J. D.
Compilers - Principles, Techniques and Tools
Addison-Wesley

[2] Advanced Compiler Design Implementation
Steven S. Muchnick
Elsevier, 2003

[3] Compiler Design in C, Allen Holub

[4] Modern compiler implementation in C by Appel

[5] System Software: An Introduction to Systems Programming
Leland L. Beck, Addison-Wesley.

[6] Systems Programming, John J. Donovan,
McGraw - Hill Education

Evaluation Policy

EndSem – 40%

MidSem – 30%

Quizzes, Internal Assessments – 30%